MA 587 Homework 02

Due: Friday 11:59pm, March 06, 2026

It should be submitted through Gradescope. PLEASE ASSIGN PAGES to each problem (making sure you're doing this correctly). PLEASE make sure the file is in CORRECTLY ORIENTED. **Show all work and you MUST write up your own solutions.**

1. Let u be twice continuously differentiable functions on $[0,1]$ with $u(0) = u(1) = 0$. Let $0 = x_0 < x_1 < \cdots < x_M < x_{M+1} = 1$ with $h_j = x_j - x_{j-1} = h$ for all j. Let $\tilde{u}_h$ be the linear interpolation of u on the uniform grid.

   (a) Show that $|u'(x) - \tilde{u}_h'(x)| \le h \max_{y\in[0,1]} |u''(y)|$. Hint: Observe that $\tilde{u}_h'(x) = \frac{u(x_j)-u(x_{j-1})}{h}$ if $x \in (x_{j-1}, x_j)$ and recall the Talyor theorem.

   (b) Let $u_h \in V_h$ be the FEM solution to the boundary value problem of
   $$-u'' = f \text{ on}(0,1) \quad u(0) = u(1) = 0.$$
   From (a), u_h satisfies $\|(u-u_h)'\| \le h \max_{y\in[0,1]} |u''(y)|$ where $\|u\| = \sqrt{\int_0^1 u^2 dx}$. Show that
   $$|u(x) - u_h(x)| \le h \cdot \max_{y\in[0,1]} |u''(y)|.$$
   Hint: $\int_0^x (u-u_h)'(s)ds = (u(x)-u_h(x)) - (u(0)-u_h(0))$ and recall Cauchy-Schwartz inequality.

2. Let V be a Hilbert space with the inner product $(\cdot,\cdot)_V$ and the norm $\|\cdot\|_V$. Let $a : V \times V \to \mathbb{R}$ be a bilinear form and $L : V \to \mathbb{R}$ be a linear form. Suppose that a is symmetric, continuous and coercive, and L is continuous. Consider
   $$\begin{aligned} &\text{(M)} \quad \text{Find } u \in V \text{ such that } F(u) \le F(v) = \frac{1}{2}a(v,v) - L(v) \quad \forall v \in V, \\ &\text{(V)} \quad \text{Find } u \in V \text{ such that } a(u,v) = L(v), \quad \forall v \in V. \end{aligned}$$

   (a) Show that (M) and (V) are equivalent.

   Let V_h be a finite dimensional subspace of V. Let u_h be the solution to the variational problem on V_h, i.e., $u_h \in V_h$ satisfies $a(u_h, w) = L(w)$ for all $w \in V_h$.

   (b) Show that
   $$\|u - u_h\|_a \le \|u - v\|_a \quad \forall v \in V_h,$$
   where $\|\cdot\|_a$ is the energy norm.

3. (2D FEM element basis)

   (a) Let $N_1, N_2, N_3 \in \mathbb{R}^2$ be three points (or nodes). Find a condition that N_1, N_2, N_3 form a triangle.

(b) Let $N_1 = (0,0), N_2 = (h,0), N_3 = (0,h)$ which form a triangle K. Construct a linear function ϕ_j satisfying $\phi_j(N_i) = 1$ if $i = j$ and 0 if $i \neq j$. Also, show that ϕ_1, ϕ_2, ϕ_3 are basis functions for $P_1(K)$. That is, any linear function on K can be written as a linear combination of $\{\phi_j\}_{j=1}^3$.

Hint: A linear function in 2D is written as a polynomial of degree 1, i.e.,

$$\phi_j(x,y) = ax + by + c.$$

(c) Let $a(u,v) = \int_K \nabla u \cdot \nabla v dx$ where dx represents the area integral. Let A be a stiff matrix whose (i,j)-component is $a(\phi_i, \phi_j)$ where ϕ_i's are defined from (b). Show that the element stiff matrix A is given by

$$\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}.$$

4. (Sobolev Space) For any $k \in \{0, 1, \dots\}$, let $H^k(\Omega) := \{v \mid D^\alpha v \in L_2(\Omega), \forall |\alpha| \leq k\}$ be the Sobolev space whose inner production is defined by $(u,v)_{H^k} = \sum_{|\alpha| \leq k} \langle D^\alpha u, D^\alpha k \rangle_{L_2(\Omega)}$ where $\langle u, k \rangle_{L_2(\Omega)} = \int_\Omega uv dx$. The inner-product induced norm is $\|u\|_{H^k} = \sqrt{(u,u)_{H^k}}$.

   (a) For any $s, k \in \{0, 1, \dots\}$ such that $s \leq k$, show that $\|u\|_{H^s} \leq \|u\|_{H^k}$.

   (b) (Triangle Inequality) One important property of the norm is the triangle inequality. That is, if $\|\cdot\|$ is a norm, it should satisfy $\|u+v\| \leq \|u\| + \|v\|$ for any elements u, v. Show that $\|u+v\|_{H^k} \leq \|u\|_{H^k} + \|v\|_{H^k}$ for all $u, v \in H^k$.

   Hint 1: Observe that $(a+b)^2 \leq a^2 + 2|a||b| + b^2 = (|a| + |b|)^2$.

   Hint 2: You can use the Cauchy-Schwarz inequality in L_2, i.e., $|\langle u, v \rangle_{L_2}| \leq \|u\|_{L_2} \|v\|_{L_2}$ for all $u, v \in L_2$.

   Hint 3: Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^M$. Then, $\langle \boldsymbol{a}, \boldsymbol{b} \rangle \leq \|\boldsymbol{a}\|_2 \|\boldsymbol{b}\|_2$. Imaging $\boldsymbol{a} = (D^\alpha u)_\alpha$. What is $\|\boldsymbol{a}\|_2$?